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MODUL 13

Laravel : Database 2

1. Setup Awal & Session-based Auth

- a. Code (boleh copy/screenshot dengan syarat terlihat jelas dan tidak blur)

Membuat model

Product.php

```
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Product extends Model
{
    use HasFactory;

    protected $fillable = ['name', 'price'];

    public function variants() {
        return $this->hasMany(Variant::class);
    }
}
```

ProductController.php

```
<?php

namespace App\Http\Controllers;

use App\Models\Product;
use Illuminate\Http\Request;
```

```
class ProductController extends Controller
{
    public function index()
    {
        $products = Product::all();
        return view('products.index', ['products' =>
$products]);
    }

    public function create()
    {
        return view('products.form', [
            'title' => 'Tambah',
            'product' => new Product(),
            'route' => route('product.store'),
            'method' => 'POST',
        ]);
    }

    public function store(Request $request)
    {
        $validated = $request->validate([
            'name' => 'required|min:4',
            'price' => 'required|integer|min:1000000',
        ]);

        Product::create($validated);
        return
redirect()->route('product.index')->with('success', 'Produk
berhasil ditambahkan');
    }

    public function show(Product $product)
    {
        return view('products.show', compact('product'));
    }

    public function edit(Product $product)
```

```

    {
        return view('products.form', [
            'title' => 'Edit',
            'product' => $product,
            'route' => route('product.update', $product),
            'method' => 'PUT',
        ]);
    }

    public function update(Request $request, Product
$product)
    {
        $validated = $request->validate([
            'name' => 'required|min:4',
            'price' => 'required|integer|min:1000000',
        ]);

        $product->update($validated);
        return
redirect()->route('product.index')->with('success', 'Produk
berhasil diperbarui');
    }

    public function destroy(Product $product)
    {
        $product->delete();
        return
redirect()->route('product.index')->with('success', 'Produk
berhasil dihapus');
    }

    public function storeVariant(Request $request, Product
$product)
    {
        $validated = $request->validate([
            'name' => 'required|min:3',
            'description' => 'required',
            'processor' => 'required',
            'memory' => 'required',

```

```

        'storage' => 'required',
    ]);

    $product->variants()->create($validated);

    return
    redirect()->route('product.index')->with('success', 'Variant
    berhasil ditambahkan ke produk ' . $product->name);
    }
}

```

Mengonfigurasi database ecommerce di modul-13/.env

```

DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=ecommerce
DB_USERNAME=root
DB_PASSWORD=

```

Menambahkan dan menjalankan seeder User di DatabaseSeeder.php

```

<?php

namespace Database\Seeders;

use Illuminate\Database\Console\Seeds\WithoutModelEvents;
use Illuminate\Database\Seeder;

class DatabaseSeeder extends Seeder
{
    /**
     * Seed the application's database.
     *
     * @return void
     */
    public function run()
    {

```

```

        if (!\App\Models\User::where('email',
'john@example.com')->exists()) {
            \App\Models\User::create([
                'name' => 'John Doe',
                'email' => 'john@example.com',
                'password' => bcrypt('password'),
            ]);
        }
    }
}

```

Membuat SiteController.php dengan autentikasi Session manual dan merancang halaman login.blade.php yang premium.

SiteController.php

```

<?php

namespace App\Http\Controllers;

use Illuminate\Http\Request;
use Illuminate\Support\Facades\Auth;

class SiteController extends Controller
{
    public function auth(Request $req)
    {
        if (Auth::attempt(['email' => $req->em, 'password' =>
$req->pwd])) {
            // jika ada nilai lain selain data user yang
ingin disimpan di session, baru gunakan session disini
            return redirect('/product');
        }

        return redirect('/login')->with('msg', 'Email /
password salah');
    }
}

```

Login.blade.php

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Login - E-Commerce</title>
    <!-- Outfit Google Font -->
    <link
href="https://fonts.googleapis.com/css2?family=Outfit:wght@30
0;400;600;800&display=swap" rel="stylesheet">
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/b
ootstrap.min.css" rel="stylesheet">
    <style>
        body {
            font-family: 'Outfit', sans-serif;
            background: linear-gradient(135deg, #0f172a 0%,
#1e293b 100%);
            min-height: 100vh;
            display: flex;
            align-items: center;
            justify-content: center;
            color: #f8fafc;
        }
        .login-card {
            background: rgba(30, 41, 59, 0.7);
            backdrop-filter: blur(16px);
            border: 1px solid rgba(255, 255, 255, 0.1);
            border-radius: 20px;
            padding: 40px;
            box-shadow: 0 10px 30px rgba(0, 0, 0, 0.3);
            width: 100%;
            max-width: 420px;
            transition: transform 0.3s ease;
        }
        .login-card:hover {
            transform: translateY(-5px);
        }
    </style>
</head>
<body>
    <div class="login-card">
        <div class="text-center">
            <h1>E-Commerce</h1>
            <h2>Login</h2>
        </div>
        <div class="text-center">
            <input type="text" class="form-control" value="Email Address" />
            <input type="password" class="form-control" value="Password" />
            <button class="btn btn-primary" type="button">Login</button>
        </div>
    </div>
</body>
</html>
```

```
}
.form-control {
    background: rgba(15, 23, 42, 0.6);
    border: 1px solid rgba(255, 255, 255, 0.1);
    color: #f8fafc;
    border-radius: 10px;
    padding: 12px;
}
.form-control:focus {
    background: rgba(15, 23, 42, 0.8);
    border-color: #3b82f6;
    color: #f8fafc;
    box-shadow: 0 0 0 3px rgba(59, 130, 246, 0.3);
}
.btn-primary {
    background: linear-gradient(135deg, #3b82f6 0%,
#2563eb 100%);
    border: none;
    border-radius: 10px;
    padding: 12px;
    font-weight: 600;
    transition: all 0.3s ease;
}
.btn-primary:hover {
    background: linear-gradient(135deg, #2563eb 0%,
#1d4ed8 100%);
    box-shadow: 0 0 15px rgba(37, 99, 235, 0.4);
    transform: scale(1.02);
}
.title-glow {
    text-align: center;
    font-weight: 800;
    background: linear-gradient(to right, #3b82f6,
#60a5fa);
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
    margin-bottom: 30px;
}
</style>
```

```

</head>
<body>
  <div class="login-card">
    <h2 class="title-glow">Sign In</h2>

    @if(session('msg'))
      <div class="alert alert-danger border-0 bg-danger
text-white bg-opacity-75 mb-4" role="alert">
        {{ session('msg') }}
      </div>
    @endif

    <form action="/login" method="POST">
      @csrf
      <div class="mb-3">
        <label for="em" class="form-label
text-secondary">Email Address</label>
        <input type="email" name="em" id="em"
class="form-control" placeholder="name@example.com" required
autocomplete="email">
      </div>

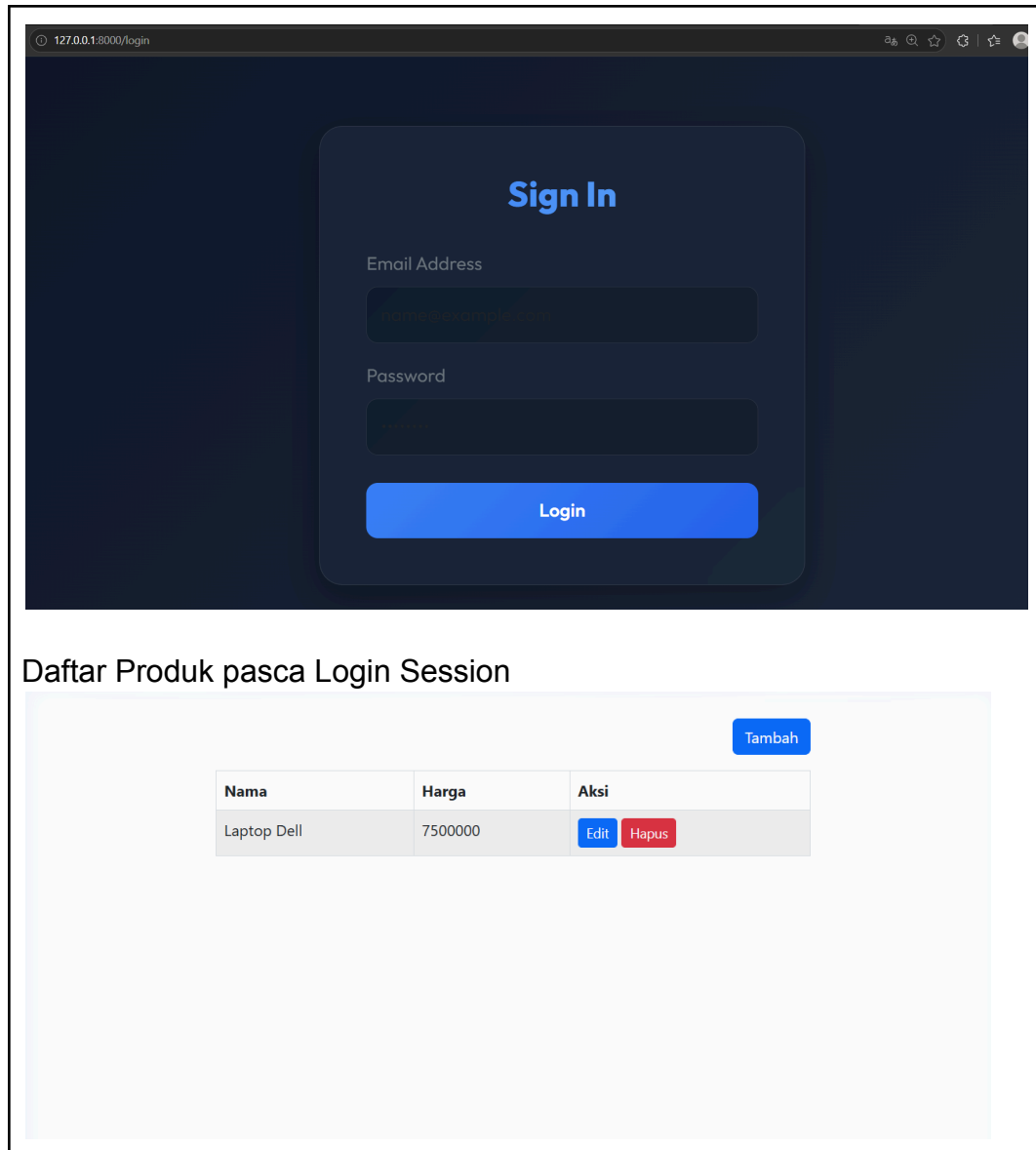
      <div class="mb-4">
        <label for="pwd" class="form-label
text-secondary">Password</label>
        <input type="password" name="pwd" id="pwd"
class="form-control" placeholder="....." required>
      </div>

      <button type="submit" class="btn btn-primary
w-100">Login</button>
    </form>
  </div>
</body>
</html>

```

b. *Screenshot output*

(Form Login)



c. Penjelasan dari code dan output (korelasi)

2. Library Auth & Middleware

a. Code (boleh copy/screenshot dengan syarat terlihat jelas dan tidak blur)

Mengubah routing di web.php untuk menggunakan Middleware auth bawaan Laravel.

```
<?php
```

```
use Illuminate\Support\Facades\Route;
use App\Http\Controllers\ProductController;
use App\Http\Controllers\SiteController;
use Illuminate\Support\Facades\Auth;

/*
| -----
|
| Web Routes
| -----
*/

Route::get('/', function () {
    return redirect('/login');
});

Route::get('/login', function () {
    if (Auth::check()) return redirect('/product');
    return view('login');
})->name('login');

Route::post('/login', [SiteController::class, 'auth']);

Route::get('/logout', function () {
    Auth::logout();
    return redirect('/login');
});

Route::post('/product/{product}/variant',
[ProductController::class,
'storeVariant'])->name('product.variant.store')->middleware('auth');

Route::resource('product',
ProductController::class)->middleware('auth');
```

Memperbarui SiteController.php untuk menggunakan Auth::attempt().

```
<?php

namespace App\Http\Controllers;

use Illuminate\Http\Request;
use Illuminate\Support\Facades\Auth;

class SiteController extends Controller
{
    public function auth(Request $req)
    {
        if (Auth::attempt(['email' => $req->em, 'password' =>
$req->pwd])) {
            // jika ada nilai lain selain data user yang
            ingin disimpan di session, baru gunakan session disini
            return redirect('/product');
        }

        return redirect('/login')->with('msg', 'Email /
password salah');
    }
}
```

Mengubah layout template.blade.php agar menampilkan info login Auth::user() dan tombol Logout.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>@yield('title') - E-Commerce</title>
    <!-- Outfit Google Font -->
    <link
href="https://fonts.googleapis.com/css2?family=Outfit:wght@30
0;400;600;700&display=swap" rel="stylesheet">
```

```
<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/b
ootstrap.min.css" rel="stylesheet">

<style>
  body {
    font-family: 'Outfit', sans-serif;
    background-color: #0f172a;
    color: #f8fafc;
    min-height: 100vh;
  }
  .navbar-custom {
    background: rgba(30, 41, 59, 0.7);
    backdrop-filter: blur(12px);
    border-bottom: 1px solid rgba(255, 255, 255,
0.1);

    padding: 15px 20px;
    border-radius: 12px;
    margin-top: 15px;
    margin-bottom: 30px;
  }
  .card-custom {
    background: rgba(30, 41, 59, 0.5);
    backdrop-filter: blur(8px);
    border: 1px solid rgba(255, 255, 255, 0.1);
    border-radius: 16px;
    color: #f8fafc;
    padding: 25px;
  }
  .table {
    color: #f8fafc;
    border-color: rgba(255, 255, 255, 0.1);
  }
  .table-striped>tbody>tr:nth-of-type(odd) {
    background-color: rgba(255, 255, 255, 0.02);
    color: #f8fafc;
  }
  .table-striped>tbody>tr:nth-of-type(even) {
    background-color: rgba(255, 255, 255, 0.05);
    color: #f8fafc;
  }
```

```

    }
    .table-bordered th, .table-bordered td {
        border: 1px solid rgba(255, 255, 255, 0.1);
    }
    .btn-warning-custom {
        background: linear-gradient(135deg, #f59e0b 0%,
#d97706 100%);
        border: none;
        color: #0f172a;
        font-weight: 600;
        padding: 8px 16px;
        border-radius: 8px;
        transition: all 0.3s ease;
    }
    .btn-warning-custom:hover {
        background: linear-gradient(135deg, #d97706 0%,
#b45309 100%);
        box-shadow: 0 0 10px rgba(245, 158, 11, 0.4);
        color: #0f172a;
    }
</style>
</head>
<body class="container">
    @auth
        <div class="navbar-custom d-flex justify-content-between
align-items-center">
            <h4 class="mb-0 text-primary fw-bold"
style="background: linear-gradient(to right, #3b82f6,
#60a5fa); -webkit-background-clip: text;
-webkit-text-fill-color: transparent;">E-Commerce Admin</h4>
            <div class="d-flex align-items-center gap-3">
                <span class="text-secondary">Logged in as:
<strong class="text-white">{{ Auth::user()->name
}}</strong></span>
                <a href="/logout" class="btn btn-warning-custom
btn-sm">Logout</a>
            </div>
        </div>
    @endauth

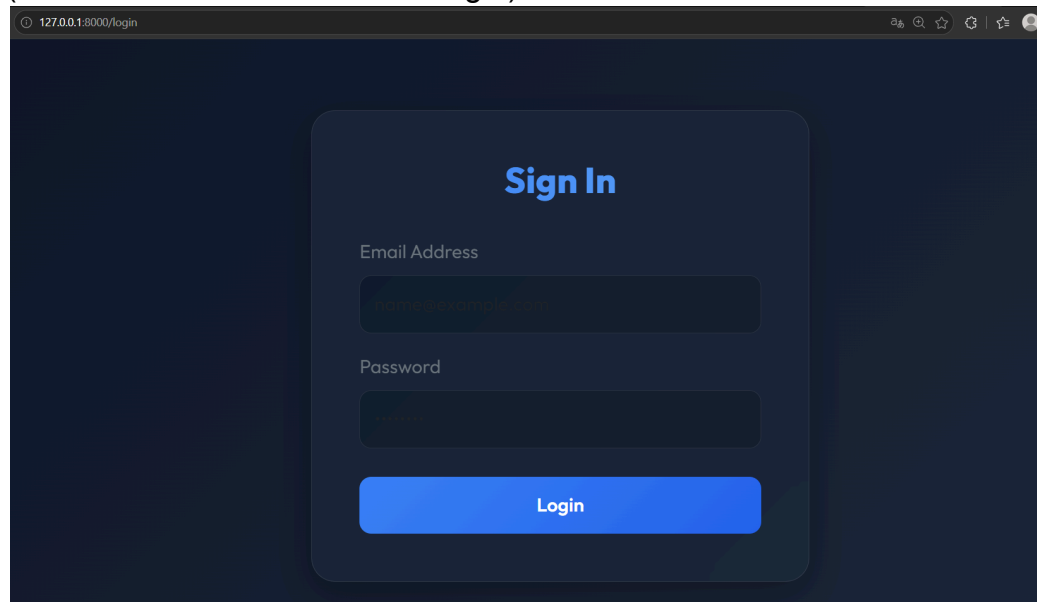
```

```
<div class="row justify-content-center">
  @yield('content')
</div>

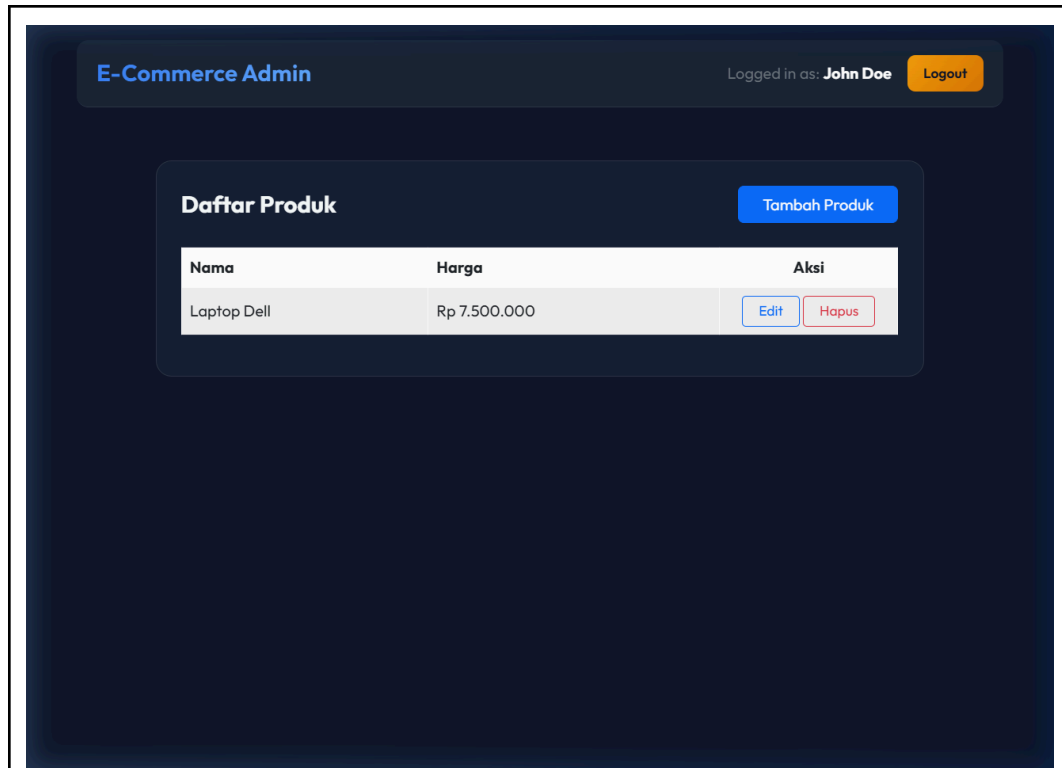
<!-- Bootstrap Bundle JS -->
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/boo
tstrap.bundle.min.js"></script>
</body>
</html>
```

b. *Screenshot output*

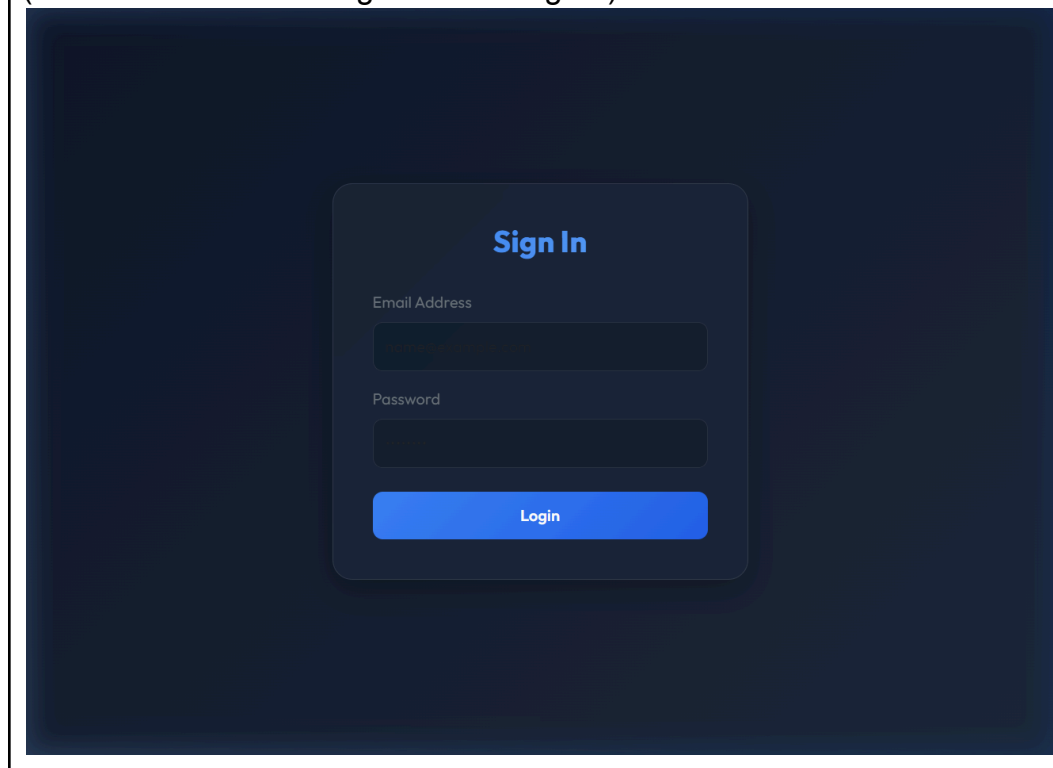
(Redirect otomatis saat belum login)



(Halaman produk dengan nama user terautentikasi)



(Kembali ke halaman login setelah logout)



c. Penjelasan dari code dan output (korelasi)

3. Model Relasi & Form Variant

- a. Code (boleh copy/screenshot dengan syarat terlihat jelas dan tidak blur)

Membuat model Variant.php dan file migrasi variant untuk mendefinisikan hubungan One-to-Many (belongsTo di Variant dan hasMany di Product).

```
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Variant extends Model
{
    use HasFactory;

    protected $fillable = ['name', 'description',
'processor', 'memory', 'storage', 'product_id'];

    public function product() {
        return $this->belongsTo(Product::class);
    }
}
```

Menambahkan kolom Variant & Spesifikasi pada tabel index.blade.php.

```
@extends('template')

@section('title', 'Daftar Produk')

@section('content')
<div class="col-md-11 mt-4">
    <div class="card-custom">
        @if(session('success'))
            <div class="alert alert-success border-0
bg-success text-white bg-opacity-75 mb-4">{{
session('success') }}</div>
        @endif
    </div>
</div>
```

```

@endif
<div class="d-flex justify-content-between
align-items-center mb-4">
    <h4 class="mb-0 fw-bold">Daftar Produk</h4>
    <a href="{{ route('product.create') }}"
class="btn btn-primary px-4">Tambah Produk</a>
</div>
<table class="table table-bordered table-striped
align-middle">
    <thead>
        <tr>
            <th>Nama</th>
            <th>Harga</th>
            <th>Variant & Spesifikasi</th>
            <th style="width: 120px;"
class="text-center">Aksi</th>
        </tr>
    </thead>
    <tbody>
        @forelse($products as $product)
            <tr>
                <td class="fw-bold">{{ $product->name
}}</td>
                <td>Rp {{ number_format($product->price,
0, ',', '.') }}</td>
                <td>
                    @if($product->variants->isNotEmpty())
                        <ul class="mb-2 text-start"
style="padding-left: 20px;">
                            @foreach($product->variants
as $var)
                                <li class="mb-2">
                                    <span
class="text-info fw-bold">{{ $var->name }}</span><br>
                                    <small
class="text-secondary">
                                        Desc: {{
$var->description }} <br />
                                        Proc: {{

```

```

$var->processor }} | RAM: {{ $var->memory }} | Strg: {{
$var->storage }} <br />

                                Product: <span
class="text-warning">{{ $var->product->name }}</span>
                                </small>
                                </li>
                                @endforeach
                                </ul>
                                @else
                                <span class="text-secondary
d-block mb-2"><small>Belum ada variant.</small></span>
                                @endif

                                <!-- Collapsible Form to Add Variant
-->

                                <button class="btn btn-sm
btn-outline-success py-1 px-3" type="button"
data-bs-toggle="collapse" data-bs-target="#addVar-{{
$product->id }}">

                                + Tambah Variant
                                </button>
                                <div class="collapse mt-3"
id="addVar-{{ $product->id }}">
                                <form action="{{
route('product.variant.store', $product->id) }}"
method="POST" class="border p-3 rounded text-start"
style="background: rgba(15, 23, 42, 0.4); border-color:
rgba(255, 255, 255, 0.1) !important;">
                                @csrf
                                <h6 class="mb-3 fw-bold
text-success">Tambah Variant Baru</h6>
                                <div class="mb-2">
                                <input type="text"
name="name" class="form-control form-control-sm text-white
bg-dark bg-opacity-50 border-secondary" placeholder="Nama
Variant (e.g. Ryzen Edition)" required>
                                </div>
                                <div class="mb-2">
                                <textarea

```

```

name="description" class="form-control form-control-sm
text-white bg-dark bg-opacity-50 border-secondary"
placeholder="Deskripsi Singkat" rows="2" required></textarea>
</div>
<div class="row g-2 mb-2">
  <div class="col-md-4">
    <input type="text"
name="processor" class="form-control form-control-sm
text-white bg-dark bg-opacity-50 border-secondary"
placeholder="Processor" required>
    </div>
    <div class="col-md-4">
      <input type="text"
name="memory" class="form-control form-control-sm text-white
bg-dark bg-opacity-50 border-secondary" placeholder="RAM"
required>
      </div>
      <div class="col-md-4">
        <input type="text"
name="storage" class="form-control form-control-sm text-white
bg-dark bg-opacity-50 border-secondary" placeholder="Storage"
required>
        </div>
      </div>
      <button type="submit"
class="btn btn-sm btn-success px-4 mt-2">Simpan
Variant</button>
    </form>
  </div>
</td>
<td class="text-center">
  <div class="d-flex flex-column gap-2
justify-content-center">
    <a href="{ route('product.edit',
$product->id) }" class="btn btn-sm
btn-outline-primary">Edit</a>
    <form method="POST" action="{
route('product.destroy', $product->id) }"
style="display:inline" onsubmit="return confirm('Yakin

```

```

hapus?') ">
                                @csrf
                                @method('DELETE')
                                <button class="btn btn-sm
btn-outline-danger w-100">Hapus</button>
                                </form>
                            </div>
                        </td>
                    </tr>
                    @empty
                    <tr>
                        <td colspan="4" class="text-center
text-secondary py-4">Belum ada data produk.</td>
                    </tr>
                    @endforelse
                </tbody>
            </table>
        </div>
    </div>
@endsection

```

Membuat form tambah variant yang interaktif (collapsible/accordion) langsung di dalam tabel produk agar mudah diuji.

Menambahkan method storeVariant di ProductController.php untuk memproses penyimpanan variant ke database.

```

<?php

namespace App\Http\Controllers;

use App\Models\Product;
use Illuminate\Http\Request;

class ProductController extends Controller
{
    public function index()

```

```

    {
        $products = Product::all();
        return view('products.index', ['products' =>
$products]);
    }

    public function create()
    {
        return view('products.form', [
            'title' => 'Tambah',
            'product' => new Product(),
            'route' => route('product.store'),
            'method' => 'POST',
        ]);
    }

    public function store(Request $request)
    {
        $validated = $request->validate([
            'name' => 'required|min:4',
            'price' => 'required|integer|min:1000000',
        ]);

        Product::create($validated);
        return
redirect()->route('product.index')->with('success', 'Produk
berhasil ditambahkan');
    }

    public function show(Product $product)
    {
        return view('products.show', compact('product'));
    }

    public function edit(Product $product)
    {
        return view('products.form', [
            'title' => 'Edit',
            'product' => $product,

```

```

        'route' => route('product.update', $product),
        'method' => 'PUT',
    ]);
}

public function update(Request $request, Product
$product)
{
    $validated = $request->validate([
        'name' => 'required|min:4',
        'price' => 'required|integer|min:1000000',
    ]);

    $product->update($validated);
    return
    redirect()->route('product.index')->with('success', 'Produk
    berhasil diperbarui');
}

public function destroy(Product $product)
{
    $product->delete();
    return
    redirect()->route('product.index')->with('success', 'Produk
    berhasil dihapus');
}

public function storeVariant(Request $request, Product
$product)
{
    $validated = $request->validate([
        'name' => 'required|min:3',
        'description' => 'required',
        'processor' => 'required',
        'memory' => 'required',
        'storage' => 'required',
    ]);

    $product->variants()->create($validated);
}

```

```

        return
    redirect()->route('product.index')->with('success', 'Variant
    berhasil ditambahkan ke produk ' . $product->name);
    }
}

```

b. *Screenshot output*

(Menampilkan detail variant yang ditambahkan beserta relasi ke produk induknya)

The screenshot shows the 'E-Commerce Admin' dashboard. At the top, it says 'Logged in as: John Doe' with a 'Logout' button. A green success message banner reads: 'Variant berhasil ditambahkan ke produk Asus ROG Strix'. Below this is the 'Daftar Produk' section with a 'Tambah Produk' button. The main content is a table with the following data:

Nama	Harga	Variant & Spesifikasi	Aksi
Laptop Dell	Rp 7.500.000	Belum ada variant. + Tambah Variant	Edit Hapus
Asus ROG Strix	Rp 18.500.000	Scar 15 Desc: High-end gaming variant with RGB lightbar Proc: AMD Ryzen 9 6900HX RAM: 32GB DDR5 Strg: 2TB PCIe Gen4 SSD Product: Asus ROG Strix + Tambah Variant	Edit Hapus

c. Penjelasan dari code dan output (korelasi)